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# Intermittent air mixing system for anaerobic digestion of animal wastewater: Operating conditions and full-scale validation

Hongnan Yang [a,b], Liangwei Deng [a,b,*], Jianwang Wu [c], Wenguo Wang [a,b], Dan Zheng [a,b], Zhiyong Wang [a,b], Yi Liu [a,b]

[a] Biogas Institute of Ministry of Agriculture and Rural Affairs, Chengdu 610041, PR China
[b] Key Laboratory of Development and Application of Rural Renewable Energy, Ministry of Agriculture and Rural Affairs, Chengdu 610041, PR China
[c] Shijiazhuang Chengding Environmental Protection Technology Co. Ltd., Shijiazhuang 050000, PR China

## HIGHLIGHTS

- Operating condition of air mixing for anaerobic digestion was investigated.
- The maximum air mixing time was determined.
- A full-scale digester was selected to investigate the feasibility of air mixing.
- The air mixing was applied in full-scale plant successfully.

## ARTICLE INFO

*Keywords:*
Air mixing
Anaerobic digestion
Effective methane production time
Mixing time
Swine wastewater

## ABSTRACT

An air mixing system for anaerobic digestion has been proved to be beneficial for methane production. The aim of the present study was to further investigate the appropriate conditions for air mixing. The effective methane production time (EMPT) was defined to determine the air mixing time in the article. The results indicated that the appropriate aeration intensity was 66.7 mL air per volume of reactor per min and mixing time was 1.5 min. When air mixing time exceeded 3 min on each occasion, total $CH_4$ production was less than that achieved under the no mixing condition due to a decrease in the EMPT. In addition, the possibility of air mixing was evaluated in an anaerobic full-scale plant comprising a continuous stirred tank reactor. One year of operating data validated the feasibility of air mixing during the anaerobic digestion of swine wastewater.

## 1. Introduction

In China, large-scale and intensive livestock production results in the rapid increase of wastewater quantities in a few concentrated areas (Duan et al., 2019; Chen et al., 2020b). According to the statistical data from National Bureau of Statistics of China (NBSC) (2017), annual animal wastewater production in China is about 3.8 billion tons, inclusive of both slaughtered and breeding stocks. The improper manure management has contributed to significant environmental problems (Pantelopoulos and Aronsson, 2021; Zhuang et al., 2020). It is estimated that livestock manure is responsible for 11%, 29%, and 52% of greenhouse gas (GHG) (such as $CH_4$ and $N_2O$), and $NH_3$ emissions in China, respectively (Zhuang et al., 2019, 2020). Anaerobic digestion (AD) is the preferred method for animal wastewater treatment because of its beneficial effects in terms of waste disposal, renewable energy recovery, and greenhouse gas emission reduction (Deng et al., 2016; Tseng et al., 2019). However, the AD of animal wastewater is limited by the low efficiency of biogas production and unstable operation because of low temperatures and slow degradation kinetics of complex organics in animal wastewater (Deng et al., 2016).

Mixing plays a prominent role in determining the efficiency of the anaerobic digestion process of animal wastewater. Appropriate mixing could improve the substrate-sludge contact with slight mass transfer resistance, enhance the hydrolysis step, ensure homogeneous distribution of substrate and micro-organisms and avoid settlings, scum and floating layers (Lindmark et al., 2014; Kariyama et al., 2018; Singh et al., 2019; Wang et al., 2019). The three different types of mixing modes applied in AD comprise mechanical mixing, hydraulic mixing, and

* Corresponding author.
*E-mail address:* dengliangwei@caas.cn (L. Deng).

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biogas mixing (Kariyama et al., 2018). Many studies have investigated the impacts of different mixing modes on the biogas production efficiency (Karim et al., 2005; Kariyama et al., 2018; Singh et al., 2019). Mechanical mixing can greatly enhance the disintegration of foam (scum) because of its large action radius. However, the high power dissipation under impeller mixing is expensive compared with biogas recirculation in terms of the maintenance/operating costs (Wu, 2012; Latha et al., 2019). For example, Trad et al. (2016) compared the energy consumption rates under mechanical mixing, hydraulic mixing, and biogas mixing, where the results showed that under the same shear level, the energy consumption was lower under biogas mixing than hydraulic mixing and mechanical mixing. Hydraulic mixing is usually only suitable for small reactors with low solid concentrations considering the occurrence of short circuits and dead zones (Karim et al., 2005; Bhattacharjee et al., 2015). Biogas mixing is considered a simple and rapid alternative mixing technique for use in future AD plants because the absence of moving elements inside digesters reduces the amount of wear and maintenance (Dapelo et al., 2015; Kuczman et al., 2017). However, biogas mixing is not used frequently in on-farm biogas plants because it is difficult to maintain adequate mixing when the biogas production rate is low.

Based on previous assessments of the impacts of micro-aeration on AD treatment (Lim and Wang, 2013; Lim et al., 2014; Xu et al., 2014; Peces et al., 2016; Chen et al., 2020a), a new mixing method for AD was developed in our previous research by using air as a gas source for mixing (referred to as "air mixing") in order to solve the problems of insufficient biogas production and safety associated with biogas mixing (Yang and Deng, 2020). The feasibility of air mixing was demonstrated based on comparisons with other mixing modes (biogas mixing, mechanical mixing, and no mixing) at the same power input. The results indicated that air mixing had no negative effects on AD under limited aeration condition, and it could also improve the methane production efficiency and reduce the hydrogen sulfide ($H_2S$) content of the biogas by 69% compared with reactors without air mixing (Yang and Deng, 2020). However, the operating conditions for air mixing remain unclear and needs to be developed. Optimizing air mixing is essential for achieving effective mixing and improving the biogas quality, particularly in terms of conditions such as the aeration intensity (or air flow rate), total volume of air for mixing, and mixing time. The aeration intensity determines whether the substrate and sludge are mixed completely or effectively. The optimum volume of air avoids the negative effects of an excessive air volume, including the dilution of the methane content and the aerobic degradation of organic matter. An excessive air flow rate and mixing time will disrupt the reducing environment for AD, and decrease the growth and activity of obligate anaerobes, especially methanogens, thereby negatively affecting the AD process. In addition, it is considered that air mixing is feasible for small-scale test equipment, but it is doubtful whether air mixing can be applied to full-scale anaerobic digesters.

In the present study, in order to address the problems described above, the optimum conditions for air mixing were determined in order to develop the air mixing technique. The possibility of applying air mixing on an industrial scale was evaluated in a continuous biogas digester with a volume of 2000 $m^3$.

## 2. Materials and methods

### 2.1. Anaerobic digesters and materials

A 4.5 L laboratory-scale air mixing reactor was established (Supporting Information). The reactor had a working volume of 4.0 L and was mixed by introducing air into the reactor via a digital peristaltic pump and an aeration disc at the bottom of the reactor. The temperature was maintained at 35 ± 1°C using a water jacket and thermostat.

Swine wastewater is a typical type of animal wastewater. In this study, swine wastewater was used as the influent for a laboratory-scale experiment. The wastewater was collected from a swine farm located in Jianyang City, Sichuan Province, China. In order to avoid clogging problems in the reactor systems, large particles were sieved using a 6 mesh screen (3.35 mm) after transporting the waste to the laboratory. The sieved swine wastewater was stored at 4°C until use. The sludge inoculant for the AD experiments was harvested from a laboratory-scale mesophilic anaerobic digester used to treat swine wastewater (Supporting Information).

### 2.2. Experimental design

#### 2.2.1. Influence of aeration intensity on biogas production

At the start of the experiment, 2 L of inoculum sludge was added to air mixed reactor. The reactor was fed swine wastewater once per day (semi-continuous mode) at a loading rate of 6.0 g VS/L/d. The steady state was defined as the point when biogas production varied within 10% of the mean between the feeding times within 2–3 weeks.

The aeration intensity was lowered from 66.7 mL/L/min to 33.3 mL/L/min. The mixing times were 1.5, 3, 5, and 10 min. The biogas production rate and biogas composition, as well as ORP were measured. The experiments were repeated in triplicate.

#### 2.2.2. Influence of different mixing conditions on mass transfer

The mass transfer in a reactor was reflected by the distribution of the total solids (TS) within the reactor, which was determined by collecting samples for TS analysis from three different heights (10, 20, and 30 cm) above the base of the reactor after mixing. The coefficient of variance (κ) was defined as the ratio of the standard deviation in the TS concentrations relative to the average TS concentration, as described by Eq. (1), and it was used to measure the degree of mixing. Furthermore, the "homogeneous mixing time" was defined as the time required to achieve a certain degree of mixing. In this study, a κ-value (less than 0.02) was considered to indicate homogeneous mixing.

$$\kappa = \frac{\sqrt{\sum_{1}^{i} \frac{(C - C_{av})^2}{i}}}{C_{av}} \quad (1)$$

In Eq. (1), $i$ is the number of samples, $C$ is the TS concentration (g/L), and $C_{av}$ is the average TS concentration (g/L).

#### 2.2.3. Influence of mixing time on biogas production

The air mixed reactor was operated as described in Section 2.2.1. The mixing times were set as 0, 1, 1.5, 3, 5, 10, 60, and 120 min. A constant aeration intensity of 66.7 mL/L/min was set for different mixing times and the mixing frequency was once per day. The biogas production rate and biogas composition were measured online at intervals of 10 min after feeding. ORP was monitored each day by an on-line probe. The experiments were repeated in triplicate.

### 2.3. Full-scale validation

Air mixing was performed for full-scale AD in a biogas plant that treated 150–300 $m^3$/d of wastewater from swine farms with 7000–16,000 pigs. The biogas plant also separated the low content liquid and concentrated liquid (Deng et al., 2014, 2016; Yang et al., 2015; Liu et al., 2018). The raw swine wastewater was collected from a pig farm in a tank each day and then pumped into a sedimentation tank (see Supporting Information). Swine wastewater was separated into the concentrated liquid (50–100 $m^3$/d) and low content liquid (100–200 $m^3$/d) by gravity sedimentation. The low content liquid and concentrated liquid were anaerobically treated by an upflow anaerobic sludge blanket (UASB) and continuous stirred tank reactor (CSTR), respectively. The biogas was stored in a gasholder and used for power generation or to heat the piggery. The CSTR was mixed by air and heated by the excess heat from the biogas generator, whereas the UASB was not

mixed or heated. The digestion temperature in the CSTR was maintained at approximately 30°C. The working volume of the full-scale CSTR reactor was 2000 $m^3$, with a diameter of 16 m and height of 10 m. The biogas was stored in a double membrane gasholder with a working volume of 500 $m^3$. The daily input quantity of the concentrated liquid in the CSTR was 53–106 $m^3$ and the feed concentration was about 5% TS. The hydraulic retention time was 19–38 d. The organic loading rate of the reactor was 1.32–2.64 kg TS/$m^3$/d. Air was compressed automatically by an air compressor into a buffer tank with a volume of 1.0 $m^3$ and pressure of 0.8 MPa (Supporting Information). To mix the reactor, compressed air was released into the reactor by opening the solenoid valve in the connecting pipe between the buffer tank and the air distributor in the reactor. The mixing frequency and mixing time were 2–3 times per day and 1.5–2 min each time, respectively. The biogas plants were monitored for 12 months, where the biogas production and temperature were recorded each day.

### 2.4. Analytical methods

The Analyse of TS was performed according to standard methods (APHA, 1998). The ORP was monitored online by an ORP probe (InPro 325X probe, M300 controller, Mettler Toledo, Urdof, Switzerland). The biogas produced in the bioreactors was collected in gas-tight bags. Biogas composition was determined using a gas chromatograph (GC) (SP-2100, Beijing Beifen-Ruili Analytical Instrument Co., Ltd., China) equipped with a thermal conductivity detector (TCD) and a packed Porapak Q column (3 m × 3 mm). Helium gas was used as the carrier gas (flow rate 10 mL/min). The temperatures of the inlet, the oven and the TCD were 100, 100 and 55°C, respectively. Details of the analytical methods were described in (Yang and Deng, 2020).

### 2.5. Data analyses

The SGompertz model equation was used to fit the $CH_4$ concentration data curves and to determine the variation in the $CH_4$ content over time. The formulas are showed in Eq. (2).

$$C_{CH4} = C_{CH4max} \cdot \exp(-\exp(d \cdot (t - t_c))) \quad (2)$$

In Eq. (2), $C_{CH4}$ is the concentration of $CH_4$ (%), t is the time (min), $C_{CH4max}$ is the maximum concentration of $CH_4$ (%), $d$ is a dimensionless kinetic constant, and $t_c$ is the time required for the $CH_4$ concentration to reach half of the maximum value (min). $C_{CH4max}$, $d$ and $t_c$ of the parameters were estimated by Nonlinear Curve Fit program in OriginPro 8.6.

## 3. Results and discussion

### 3.1. Optimized conditions for air mixing in AD

The feasibility of air mixing in AD has been proved in our previous pilot study by compared the performance of AD reactors with different mixing modes (air mixing, biogas mixing, and no mixing) (Yang and Deng, 2020). The optimal conditions determined for air mixing in the AD process, such as the aeration intensity and time of air mixing, were presented in the following.

#### 3.1.1. Aeration intensity

The power consumption required for mixing is likely to influence the economic cost of the process. The power input is generally 5–10 W/$m^3$ for biogas mixing (Metcalf et al., 2003) and was altered in each experiment by varying the aeration intensity. When the aeration intensity was 66.7 mL/L/min, the corresponding mixing power input was estimated to be about 5 W/$m^3$. In order to reduce the consumption of electrical energy, the aeration intensity was reduced from 66.7 mL/L/min to 33.3 mL/L/min, which corresponded to a power input of 2.5 W/$m^3$. The

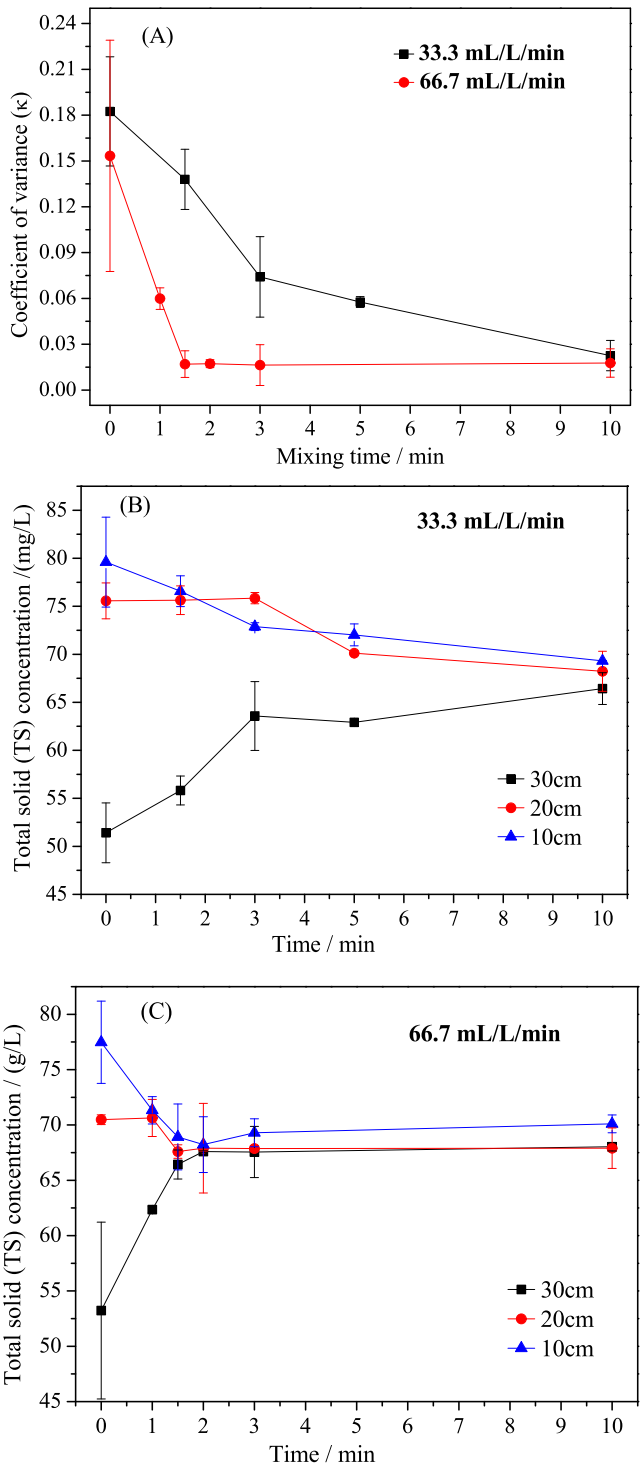



**Fig. 1.** Changes in the coefficient of variance (κ) (A) and total solids (TS) concentrations over time at mixing intensities of 33.3 mL/L/d (B) and 66.7 mL/L/d (C).

effects of the power input on the mixing degree were presented in Fig. 1. As shown in Fig. 1, the κ value was 0.138 with a low aeration intensity of 33.3 mL/L/min and mixing time of 1.5 min, which was much higher (one order of magnitude) than that (0.017) with the high aeration intensity (66.7 mL/L/min) and the same mixing time. Furthermore, the low aeration intensity resulted in the stagnation at the bottom of the reactor and near the walls, although the upper zone at the top of the reactor was well mixed due to bubbles rupturing, as shown in Fig. 1(B).

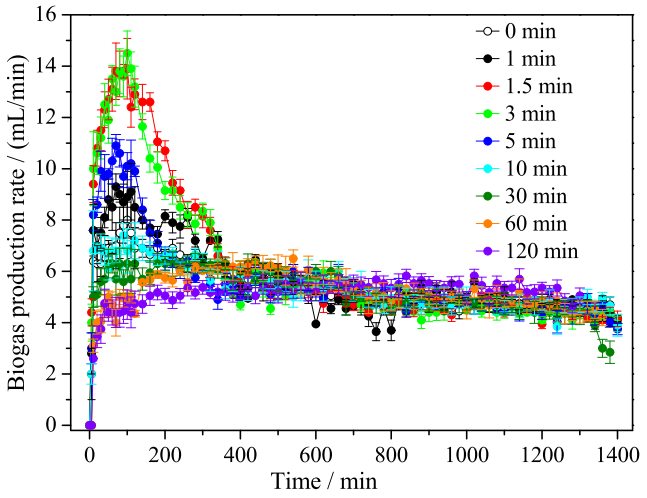



**Fig. 2.** Changes in biogas production rate over time after stopping mixing with different air mixing times.

When the mixing time increased to 10 min, homogeneous mixing was achieved (the κ value was equal to 0.023), but the power consumption was greatly increased due to the long mixing time. With a mixing time of 3 min and aeration intensity of 33.3 mL/L/min, the consumption of electrical energy was equivalent to that with the higher aeration intensity (66.7 mL/L/min) and a mixing time of 1.5 min. However, the κ value was still lower than that with the high aeration intensity (66.7 mL/L/min) and the biogas production was 7.4% lower than that with the high aeration intensity (see Supporting Information). In addition, the low aeration intensity yielded a smaller quantity of gas from the reactor and the $CH_4$ content of the biogas was slightly lower than that with the high aeration intensity. These results demonstrate that the power input for air mixing should exceed 2.5 $W/m^3$ because it was difficult to achieve homogeneous mixing under this power input even with a long mixing time.

It is expected that the percentage of dead zones will decrease and the mixing degree will increase as the aeration intensity (energy input) increases (Wu, 2010; McLeod et al., 2018). However, the results obtained in previous studies indicate that this effect is very weak (Kariyama et al., 2018). For example, increasing the gas circulation rate from 65 to 197 mL/L/min did not significantly reduce the dead zones inside a flat bottom digester according to Karim et al. (2004). In addition, excessive aeration intensity can damage microbial flocs near the exits where strong turbulent flows exist and increasing the mixing input is uneconomical (Zhang et al., 2016). The rheological properties and density of the liquid mixed should also be carefully considered when determining the appropriate aeration intensity.

### 3.1.2. *Effect of different air mixing time on biogas production*

The study of aeration intensity indicated that when the aeration intensity was 66.7 mL/L/min, the air mixing system could achieve a homogeneous mixing in a short time (far below 10 min) and accordingly low amount of air used to mixing. The air mixing time was further optimized at the aeration intensity of 66.7 mL/L/min in terms of biogas production rate, biogas composition and methane production.

*3.1.2.1. Effect on biogas production rate.* To investigate the effect of the air mixing time on biogas production, the biogas production rates (mL/min) were measured under different air mixing times after mixing was stopped, as shown in Fig. 2. The biogas production rate was higher than that under no mixing (mixing time = 0 min) when the air mixing time did not exceed 5 min and the positive impact continued for about 5–6 h. The maximum biogas production rate was about 14.0 mL/min with mixing times in the range of 1.5–3 min, which was 75.0% higher than that under 8.0 mL/min with no mixing. Our observations showed that increasing the air mixing time did not significantly improve the production of biogas. When the continuous mixing time exceeded 30 min, the biogas production rate decreased compared with that under no mixing. Similar results were obtained in previous studies of the three common types of mixing mode, where the digester was mixed well under an intermittent mixing mode with a relatively short mixing time (Kress et al., 2018; Mao et al., 2019). However, these results may have different explanations. In particular, a long air mixing time may have led to inhibition of the AD system due to the presence of air. In addition, the readily degradable organic matter may have been aerobically oxidized in the mixing stage to leave a high proportion of refractory substances, thereby reducing the gas production rate (Nguyen et al., 2019).

*3.1.2.2. Effect on biogas composition.* The variations in the $CH_4$ content were determined at different air mixing times (1, 1.5, 3, 5, 10, 30, 60, and 120 min) to estimate the appropriate air mixing time (Fig. 3). The response of the reactor to the injection of air (an oxidant) was very rapid and the $CH_4$ content dropped markedly from 58% to below 1% as the air mixing time increased from 0 to more than 15 min (Fig. 3(A)). Although the air mixing time was as long as 2 h, the $CH_4$ content gradually returned to its threshold value when air mixing stopped as shown in Fig. 3(B) and the ORP value returned to less than −400 mV (see Supporting Information), which indicated the AD systems were not disrupted. Therefore, the air mixing time and subsequent recovery time for the $CH_4$ content were key factors that determined the appropriate air mixing time.

The variation in the $CH_4$ content over time after air mixing stopped could be described by Eq. (2). The fitted results obtained for different air mixing times are shown in Table 1, where the $R^2$ values over 0.9 indicate that the fitted results were good. The recovery time required for the $CH_4$ content to recover to a certain value was calculated using Eq. (2). The threshold value for the $CH_4$ content was determined as 40%. The corresponding recovery times for different air mixing tines are shown in Table 2. The results demonstrate that the recovery time increased from 0.08 h to 3.83 h as an increase of air mixing time and it took more than 1 h to reach the threshold value of 40% when the air mixing time was more than 3 h.

*3.1.2.3. Optimum air mixing time.* The results presented above demonstrate that a long air mixing time had negative impacts on the biogas production rate and biogas composition, thereby affecting methane production. We then determined the maximum acceptable mixing time for which the methane production could not be lower than that with no mixing. Two new metrics were developed to compare the results. The first metric called the "non-effective $CH_4$ production time (NEMPT)" was defined as the time when the $CH_4$ content dropped below a specified threshold (40% in this study) after air mixing. The other metric was the converse of NEMPT, i.e., the "effective $CH_4$ production time (EMPT)," which was defined as the difference between 24 h and NEMPT. The NEMPT, EMPT, $CH_4$ production rate ($Pr_{CH4}$), and total $CH_4$ production volume ($P_{CH4}$) within EMPT under different mixing times are summarized in Table 2. The total $CH_4$ production ($P_{CH4}$) was calculated as $Pr_{CH4}$ multiplied by EMPT. As shown in Table 1, a long mixing time resulted in a shorter EMPT and the EMPT decreased greatly to only 18.17 h under continuous mixing for 2 h on each occasion. When the air mixing time was less than 3 min, $P_{CH4}$ exceeded that under no mixing, and the largest $P_{CH4}$ was obtained when the mixing time was 1.5 min. These results indicated that the maximum air mixing time should be 3 min at an aeration intensity of 66.7 mL/L/min, and the appropriate air mixing time was 1.5 min. A mixing time of 1.5 min produced the greatest amount of $CH_4$ and minimized the consumption of electrical energy to make the process more economical.

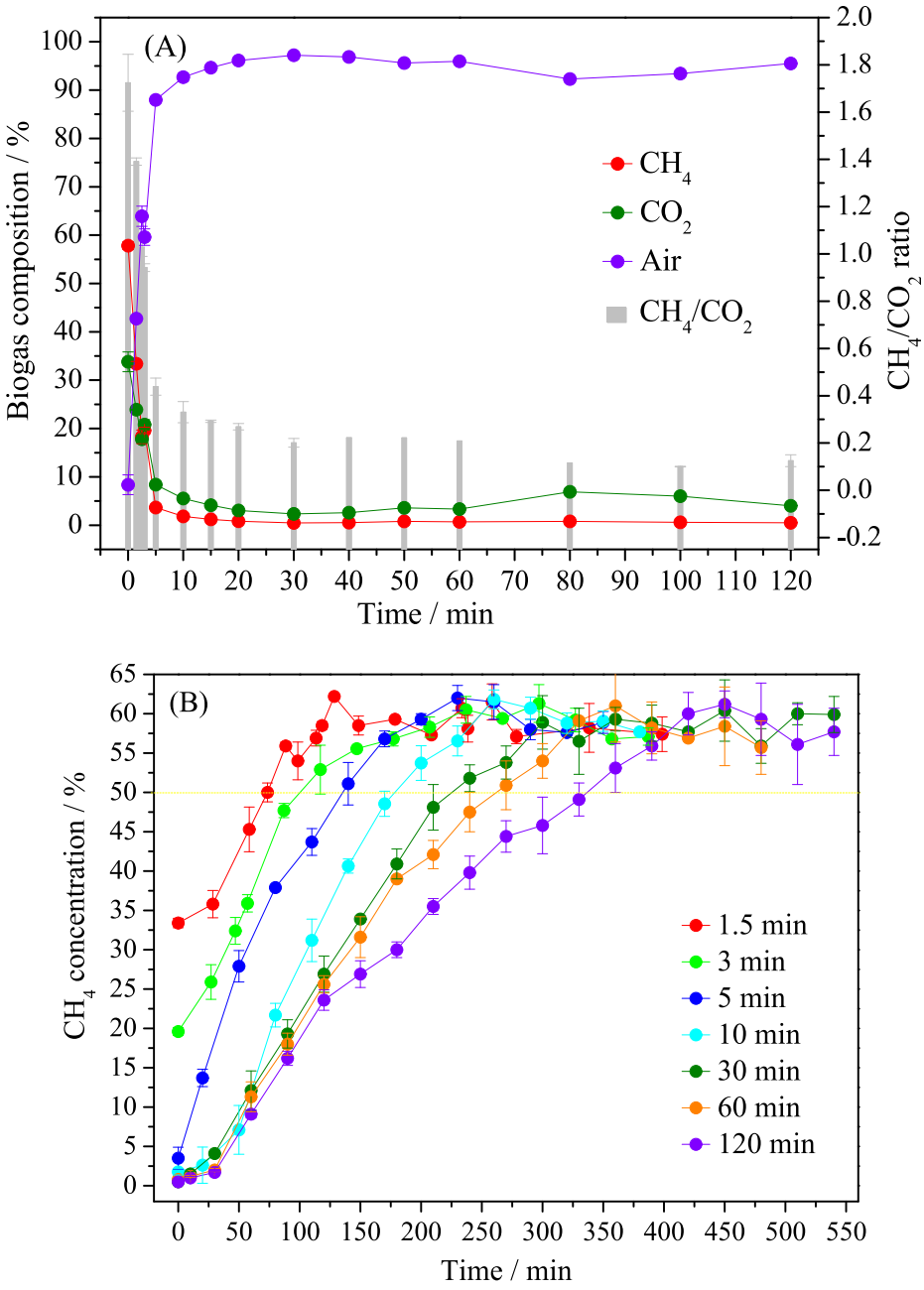



**Fig. 3.** Biogas composition during the air mixing period (A) and after stopping air mixing with different air mixing times (B).

**Table 1**
. Fitted results obtained for the $CH_4$ concentration after stopping mixing with different air mixing times.

| Mixing time/min | 1 | 1.5 | 3 | 5 | 10 | 30 | 60 | 120 |
|---|---|---|---|---|---|---|---|---|
| $C_{CH4max}$ | 59.8 | 59.6 | 59.7 | 60.1 | 60.5 | 59.9 | 59.6 | 61.3 |
| $t_c$ | −31.3 | −19.7 | 13.6 | 41.7 | 86.1 | 101.6 | 109.5 | 134.8 |
| C | 0.020 | 0.020 | 0.018 | 0.020 | 0.019 | 0.013 | 0.012 | 0.009 |
| $R^2$ | 0.914 | 0.913 | 0.979 | 0.987 | 0.994 | 0.996 | 0.992 | 0.988 |

$C_{CH4max}$: maximum concentration of $CH_4$ (%); $t_c$: time required for the $CH_4$ concentration to reach half of its maximum value (min); *d*: a dimensionless kinetic constant.

### *3.2. Validation based on a full-scale anaerobic digester*

Air mixing was applied in a full-scale anaerobic digester for treating swine wastewater to validate further its feasibility. The aeration intensity and mixing time were set as 3.0–4.5 mL/L/min and 1.5–2 min, respectively, in the full-scale anaerobic digester (see Section 2.3). The aeration intensity approached the limit of 5–7 mL/L/min recommended by Metcalf et al. (2003) for biogas mixing and its corresponding power input was 3.5–5 $W/m^3$. These design parameters closed to optimum condition obtained from the laboratory scale experiment as described in Section 3.1 that the power input for air mixing should exceed 2.5 $W/m^3$ in view of degree of mixing and mixing performance, and the air mixing

**Table 2**
. Effective $CH_4$ production time (EMPT) and $CH_4$ production rate ($P_{CH4}$) with different mixing times.

| Mixing time/min | 0 | 1 | 1.5 | 3 | 5 | 10 | 30 | 60 | 120 |
|---|---|---|---|---|---|---|---|---|---|
| $RT_{CH4}$ (h) | 0 | 0.08 | 0.47 | 1.07 | 1.45 | 2.22 | 2.80 | 3.05 | 3.83 |
| NEMPT (h) | 0 | 0.10 | 0.49 | 1.12 | 1.53 | 2.38 | 3.30 | 4.05 | 5.83 |
| EMPT (h) | 24 | 23.9 | 23.51 | 22.88 | 22.47 | 21.62 | 20.70 | 19.95 | 18.17 |
| $Pr_{CH4}$ (L/L/h) | 0.047 | 0.048 | 0.052 | 0.049 | 0.045 | 0.045 | 0.043 | 0.045 | 0.044 |
| $P_{CH4}$ (L/L) | 1.129 | 1.150 | 1.218 | 1.130 | 1.013 | 0.975 | 0.919 | 0.941 | 0.897 |

NEMPT: non-effective $CH_4$ production time (h) equal to the recovery time (RT) plus the air mixing time; EMPT: effective $CH_4$ production time equal to 24 h minus NEMPT; $Pr_{CH4}$: $CH_4$ production rate per h within the effective $CH_4$ generation time; $P_{CH4}$: total production of $CH_4$ in one day.

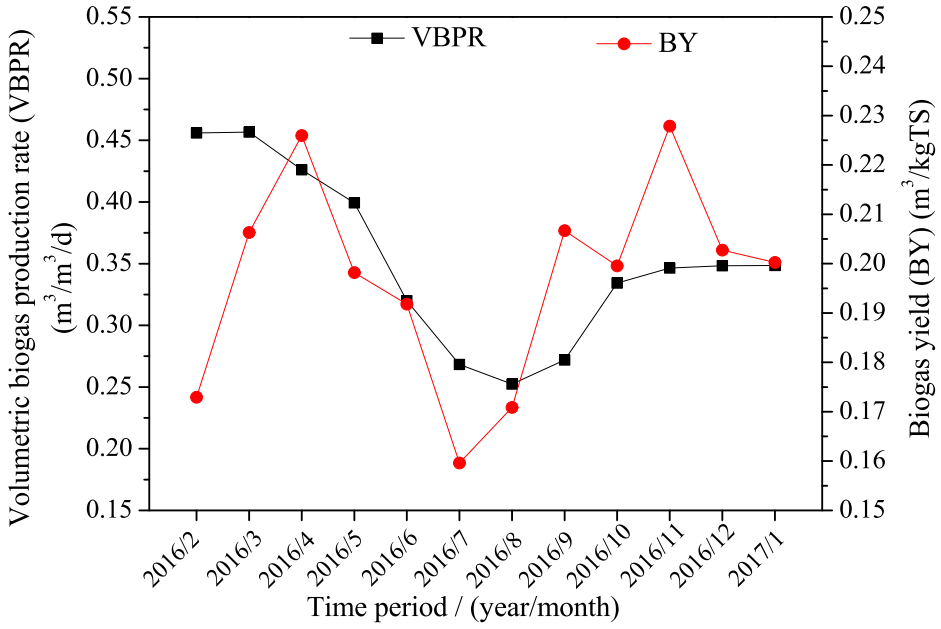



**Fig. 4.** Variations in volumetric biogas production rate and biogas yield over 1 year.

time should not exceed 3 min when the aeration intensity was 66.7 mL/L/min (the corresponding power input of 5 W/$m^3$). The full-scale anaerobic digester with air mixing system has been running steadily for 5 years. Over the past 5 years, the inlet, outlet, and slag discharge were not blocked, whereas these problems affected other biogas projects for swine wastewater treatment, thereby demonstrating that air mixing was highly effective.

The full-scale anaerobic digester was monitored for 1 year to investigate the long-term effect of air mixing on biogas production. The biogas production data obtained over 1 year are shown in Fig. 4. The volumetric biogas production rate per day varied among 0.252–0.457 $m^3/m^3/d$ and the mean value was 0.352 $m^3/m^3/d$. The biogas produced from April to September was used for power generation, and the power generation production per month was in the range of 10558–26069 kW/h. In winter (January and October to December), the biogas was used for combustion and it normally heated the boiler. The results showed that air injection did not affect the quality of the biogas. In addition, the reactor normally produced biogas, which could be used to generate electricity and as boiler fuel. Air mixing had no negative effects on biogas production and the biogas quality in AD system. In addition, due to the separation of the low content liquid and concentrated liquid, the AD fermentation tank still maintained a digestion temperature of 30°C when the temperature was below −10°C in the winter. In contrast, if the traditional process (heating all raw waste water) was adopted for swine wastewater with TS 1.0–1.5%, it was not possible to raise the digestion temperature to 30°C by using the waste heat from power generation. Thus, a biogas reactor for swine wastewater treatment could not operate normally in the winter without additional heating. Therefore, our operational performance assessment of the full-scale anaerobic reactor demonstrated that air mixing is feasible for application in the industry.

## 4. Conclusion

Operation condition in an air mixing system for anaerobic digestion was evaluated in lab-scale experiments. The appropriate aeration intensity was 66.7 mL/L/min and mixing time was 1.5 min. The air mixing time should not exceed 3 min because long mixing time led to a decrease in the effective methane production time and accordingly total $CH_4$ production. Validation of air mixing based on a full-scale anaerobic digester was conducted at the aeration intensity of as 3.0–4.5 mL/L/min and the mixing time of 1.5–2 min. Stable operation and biogas production over one year demonstrated the feasibility of air mixing.

## CRediT authorship contribution statement

**Hongnan Yang:** Methodology, Writing - original draft, Writing - review & editing. **Liangwei Deng:** Conceptualization, Resources, Writing - review & editing, Funding acquisition. **Jianwang Wu:** Validation. **Wenguo Wang:** Funding acquisition. **Dan Zheng:** Supervision, Investigation. **Zhiyong Wang:** Investigation. **Yi Liu:** Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2021.125304.

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